



amazon



SQL ANALYSIS



ABOUT COMPANY

AMAZON IS ONE OF THE WORLD'S LARGEST AND MOST CUSTOMER-CENTRIC E-COMMERCE PLATFORMS, FOUNDED BY JEFF BEZOS IN 1994. ORIGINALLY STARTING AS AN ONLINE BOOKSTORE, AMAZON HAS GROWN INTO A GLOBAL TECHNOLOGY GIANT OFFERING EVERYTHING FROM ELECTRONICS, APPAREL, AND HOME GOODS TO CLOUD COMPUTING (AWS), DIGITAL STREAMING, AND ARTIFICIAL INTELLIGENCE SERVICES.

WITH OPERATIONS IN OVER 100 COUNTRIES, AMAZON IS KNOWN FOR ITS VAST PRODUCT SELECTION, COMPETITIVE PRICING, SEAMLESS LOGISTICS NETWORK, AND CUSTOMER-FIRST APPROACH. ITS INNOVATIVE FEATURES—SUCH AS ONE-DAY DELIVERY, PERSONALIZED RECOMMENDATIONS, AND ALEXA-POWERED DEVICES—HAVE REDEFINED ONLINE SHOPPING EXPERIENCES.

AS A DATA-DRIVEN COMPANY, AMAZON LEVERAGES REAL-TIME ANALYTICS TO OPTIMIZE OPERATIONS, ENHANCE CUSTOMER SATISFACTION, AND DRIVE SALES GROWTH ACROSS MULTIPLE VERTICALS.



CORE AREAS TO FOCUS ON:

EXPLORING SALES & PRICING DATA

ANALYZE AMAZON'S PRODUCT PRICING AND DISCOUNTS TO UNDERSTAND AFFORDABILITY AND VALUE TRENDS.

SHOWCASE SQL QUERYING SKILLS

DEMONSTRATE THE USE OF SQL TO SOLVE REAL-WORLD E-COMMERCE DATA CHALLENGES EFFECTIVELY.



IDENTIFY CUSTOMER SENTIMENT

USE REVIEW CONTENT TO GAUGE CUSTOMER SATISFACTION AND HIGHLIGHT COMMON ISSUES OR PRAISES.

UNCOVER BUSINESS INSIGHTS

EXTRACT MEANINGFUL INSIGHTS RELATED TO PRODUCT PERFORMANCE, CATEGORIES, AND CUSTOMER BEHAVIOR.

Task 01: List all products with a discounted price below ₹500.



• To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT
    product_id, product_name, discounted_price
FROM
    practice.mytable
WHERE
    discounted_price < 500;
```



Output for the Query.

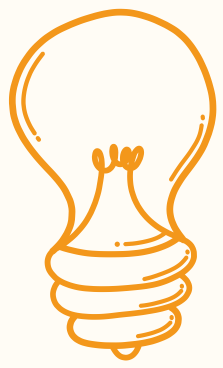


	product_id	product_name	discounted_price
▶	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Ch...	399
	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging ...	199
	B096MSW6CT	Source Fast Phone Charging Cable & Data Sync...	199
	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB ...	329
	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...	154
	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast C...	149
	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro US...	176.63
	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	229
	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), ...	499
	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging ...	199

Task 02: Find products with a discount percentage of 50% or more.



- To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT
    product_id, product_name, discount_percentage
FROM
    mytable
WHERE
    discount_percentage >= 0.50
ORDER BY discount_percentage DESC;
```



Output for the Query.

	product_id	product_name	discount_percentage
▶	B097C564GC	rts [2 Pack] Mini USB C Type C Adapter Plug, Ty...	0.94
	B0BF54972T	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch ...	0.91
	B0BF54LXW6	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch ...	0.91
	B0BF4YBLPX	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch ...	0.91
	B0BF563HB4	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch ...	0.91
	B0BF57RN3K	Fire-Boltt Ninja Call Pro Plus 1.83" Smart Watch ...	0.91
	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync...	0.9
	B085HY1DGR	Sounce Spiral Charger Cable Protector Data Ca...	0.9
	B085HY1DGR	Sounce Spiral Charger Cable Protector Data Ca...	0.9
	B0BBW521YC	LAPSTER 12pcs Spiral Cable Protectors for Char...	0.9

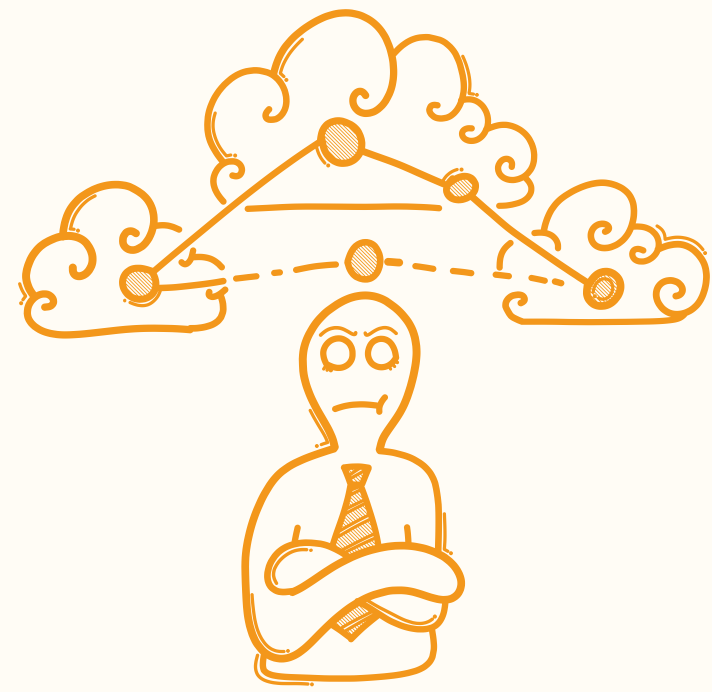
Task 03: Retrieve all products where the name contains the word "Cable."



- To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT
    product_id, product_name
FROM
    mytable
WHERE
    product_name LIKE '%Cable%';
```



Output for the Query.

	product_id	product_name
▶	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Ch...
	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging ...
	B096MSW6CT	Sounce Fast Phone Charging Cable & Data Sync...
	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB ...
	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 P...
	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast C...
	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro US...
	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)
	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging ...
	B08CF3D7QR	Portronics Konnect L POR-1081 Fast Charging 3...

Task 04: Display the difference between the average of the actual price and the average of discounted price for each product.

- To Retrieve Desired Result Use the Following Query in MYSQL:



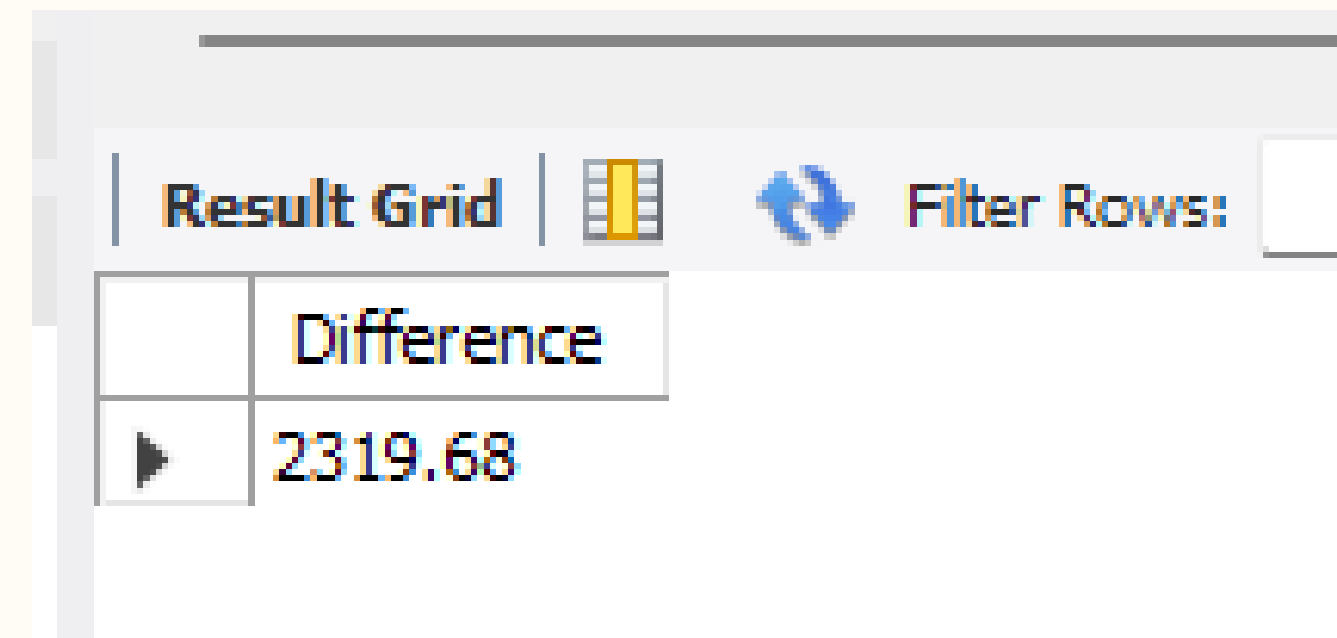
SELECT

round(**AVG**((actual_price) - (discounted_price)),2) **AS** Difference

FROM

mytable;

Output for the Query.

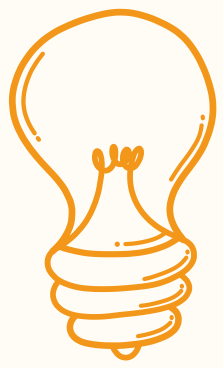


Result Grid		Filter Rows:
	Difference	
▶	2319.68	

Task 05: Query reviews that mention "fast charging" in their content.



• To Retrieve Desired Result Use the Following Query in MYSQL:

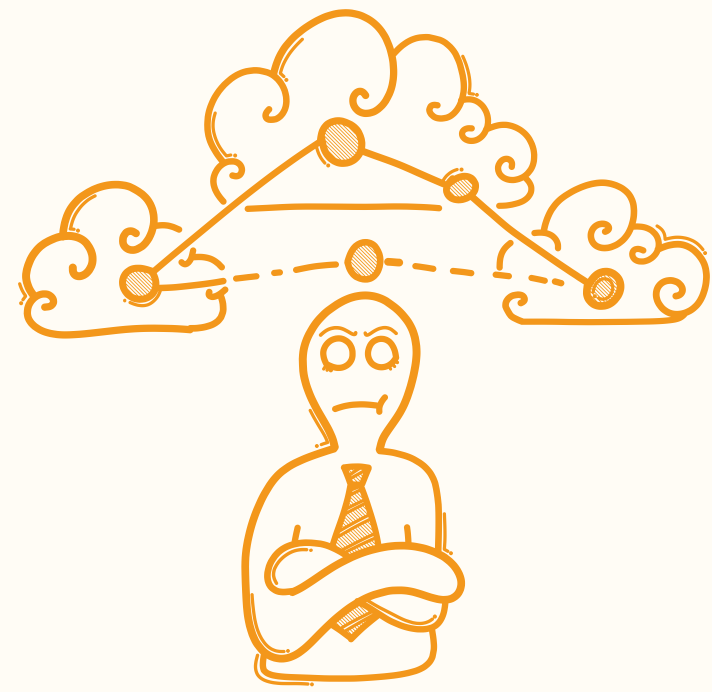


```
SELECT
    review_id, review_content
FROM
    mytable
WHERE
    review_content LIKE '%fast charging%';
```



Output for the Query.

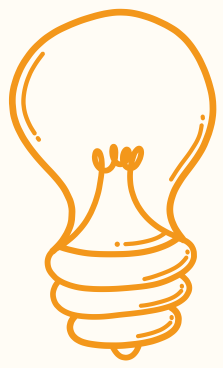
	review_id	review_content
▶	RGIQEG07...	I ordered this cable to connect my phone to An...
	R3EEUZKK...	Good product,long wire,Charges good,Nice,I bo...
	RGIQEG07...	I ordered this cable to connect my phone to An...
	R11MQS7...	The cable is efficient in fast charging but in quic...
	R3EEUZKK...	Good product,long wire,Charges good,Nice,I bo...
	R1QETDIP...	Definitely isn't as good as the original cord but ...
	R2JPQNKC...	I like it 🍌🍌,Best charging power . I used this c...
	RGIQEG07...	I ordered this cable to connect my phone to An...
	R2BP8Y5O...	Check for offera before buying,18w charging c...
	R3F4T5TR...	Worked on iPhone 7 and didn't work on iPhone ...



Task 06: Identify products with a discount percentage between 20% and 40%.

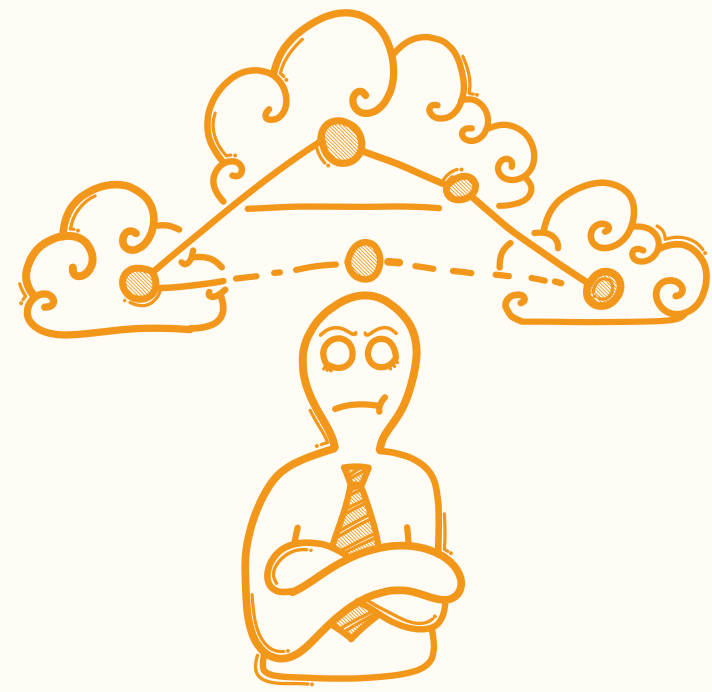


• To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT
    product_id, product_name, discount_percentage
FROM
    mytable
WHERE
    discount_percentage BETWEEN 0.20 AND 0.40;
```

Output for the Query.



Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	product_id	product_name	discount_percentage			
▶	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	0.23			
	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging ...	0.33			
	B082LSVT4B	Ambrane Unbreakable 60W / 3A Fast Charging ...	0.38			
	B08DPLCM6T	LG 80 cm (32 inches) HD Ready Smart LED TV 3...	0.39			
	B08B42LWKN	OnePlus 80 cm (32 inches) Y Series HD Ready L...	0.25			
	B094JNXNPV	Ambrane Unbreakable 3 in 1 Fast Charging Brai...	0.25			
	B0B3MMYHYW	OnePlus 126 cm (50 inches) Y Series 4K Ultra H...	0.28			
	B07KRCW6LZ	TP-Link Nano AC600 USB Wi-Fi Adapter(Archer ...	0.38			
	B014I8SSD0	Amazon Basics High-Speed HDMI Cable, 6 Feet ...	0.35			
	B09Q5SWVB1	OnePlus 80 cm (32 inches) Y Series HD Ready S...	0.27			

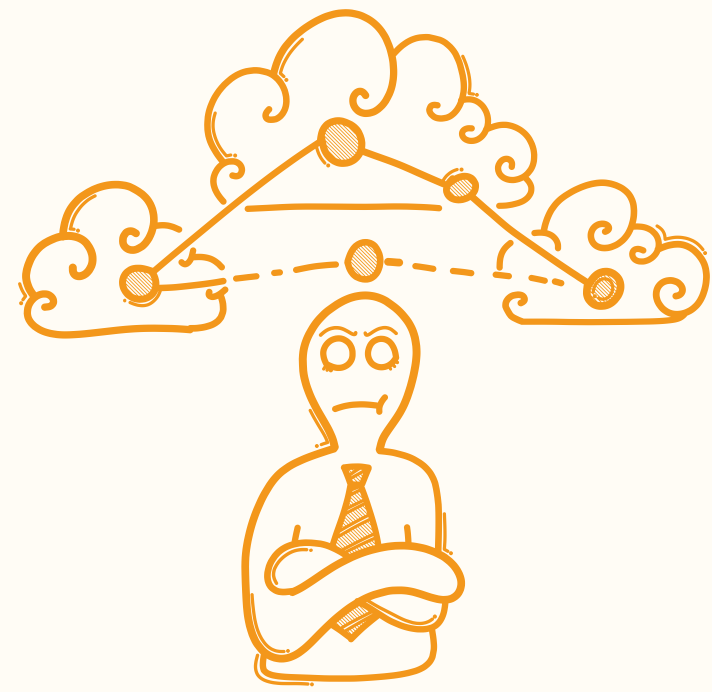
Task 07: Find products that have an actual price above ₹1,000 and are rated 4 stars or above.



• To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT
    product_id, product_name, rating, actual_price
FROM
    mytable
WHERE
    actual_price > 1000 AND rating >= 4
ORDER BY actual_price desc , rating desc;
```



Output for the Query.

Result Grid

Filter Rows:

Export:

Wrap Cell Content:

	product_id	product_name	rating	actual_price
▶	B09WN3SRC7	Sony Bravia 164 cm (65 inches) 4K Ultra HD Sm...	4.7	139900
	B0BC8BQ432	VU 164 cm (65 inches) The GloLED Series 4K Sm...	4.3	85000
	B0B3XXSB1K	LG 139 cm (55 inches) 4K Ultra HD Smart LED TV...	4.3	79990
	B09NS5TKPN	LG 1.5 Ton 5 Star AI DUAL Inverter Split AC (Co...	4.3	75990
	B08VB57558	Samsung Galaxy S20 FE 5G (Cloud Navy, 8GB R...	4.2	74999
	B0B15GSPQW	Samsung 138 cm (55 inches) Crystal 4K Neo Ser...	4.3	70900
	B09RWQ7YR6	MI 138.8 cm (55 inches) 5X Series 4K Ultra HD L...	4.3	69999
	B095JPKPH3	OnePlus 163.8 cm (65 inches) U Series 4K LED S...	4.1	69999
	B092BL5DCX	Samsung 138 cm (55 inches) Crystal 4K Series U...	4.3	69900
	B0B9XLX8VR	VU 139 cm (55 inches) The GloLED Series 4K Sm...	4.3	65000

Task 08: Find products where discounted price ends with a 9.



- To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT
    product_id, product_name, discounted_price
FROM
    mytable
WHERE
    discounted_price LIKE '%9';
```



Output for the Query.

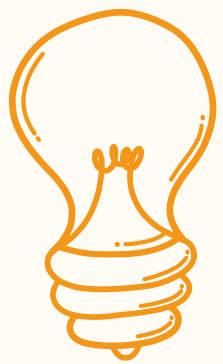


Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	product_id	product_name	discounted_price			
▶	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Ch...	399			
	B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging ...	199			
	B096MSW6CT	Source Fast Phone Charging Cable & Data Sync...	199			
	B08HDJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB ...	329			
	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast C...	149			
	B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	229			
	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), ...	499			
	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging ...	199			
	B0789LZTCJ	boAt Rugged v3 Extra Tough Unbreakable Braid...	299			
	B07KSMBL2H	AmazonBasics Flexible Premium HDMI Cable (Bla...	219			

Task 09: Display review contents that contains words like worst, waste, poor, or not good.



- To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT review_content
FROM mytable
WHERE Review_content LIKE '%worst%'
OR Review_content LIKE '%waste%'
OR Review_content LIKE '%poor%'
OR Review_content LIKE '%not good%';
```



Output for the Query.

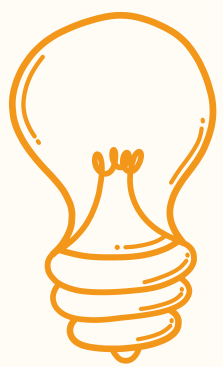


	review_content
►	Worth for money - suitable for Android auto... my purpose served in car....got it for R... Check for offera before buying,18w charging cable,,Awesome cable!Charging, data tr... LG was always Good , correct delivery, good service but sound quality was poor. The s... This connector has provided as a boon for my old Samsung LCD monitor and I am proac... Overall good.,TV picture ok smart betterSound very low another brand,Service provide... About the TV - Wonderful-----Probably... This OnePlus TV is great if you want a affordable and middle range features TV. I got it... Buy it,Received in good condition and is working properly,This price cabil not good, an... I am posting this after a few weeks of using the product.Usage: Continuous, OTT only,... NOTE:@ If you select brand-authorized TV installation, then DO NOT unbox the tv until...

Task 10: List all products where the category includes "Accessories."



- To Retrieve Desired Result Use the Following Query in MYSQL:



```
SELECT
    product_id,product_name, category
FROM
    mytable
WHERE
    category LIKE '%Accessories%';
```



Output for the Query.

	product_id	product_name	category
▶	B07JW9H4J1	Wayona Nylon Braided U...	Computers&Accessories Accessories&Per...
	B098NS6PVG	Ambrane Unbreakable 6...	Computers&Accessories Accessories&Per...
	B096MSW6CT	Source Fast Phone Char...	Computers&Accessories Accessories&Per...
	B08HDJ86NZ	boAt Deuce USB 300 2 in...	Computers&Accessories Accessories&Per...
	B08CF3B7N1	Portronics Konnect L 1.2...	Computers&Accessories Accessories&Per...
	B08Y1TFSP6	pTron Solero TB301 3A T...	Computers&Accessories Accessories&Per...
	B08WRWP...	boAt Micro USB 55 Tangl...	Computers&Accessories Accessories&Per...
	B08DDRGWTJ	MI Usb Type-C Cable Sm...	Computers&Accessories Accessories&Per...
	B008IFXQFU	TP-Link USB WiFi Adapte...	Computers&Accessories NetworkingDevic...
	B082LZGK39	Ambrane Unbreakable 6...	Computers&Accessories Accessories&Per...



KEY INSIGHTS

Extracted Subsets for Targeted Analysis

By narrowing the dataset, we created a manageable subset of products suitable for further pricing, category, and promotional evaluations.

Enabled Data Segmentation

Segmented the product data based on price thresholds, laying the groundwork for deeper comparative insights across different pricing tiers.

Established Query Framework

Demonstrated a repeatable method for querying specific conditions from a larger dataset, supporting structured and scalable analysis.



Prepared Data for Visualization or Reporting

Structured the output to be easily visualized or reported in dashboards, supporting future integration into decision-making tools.

Applied Conditional Filtering

We used SQL queries to filter the dataset and isolate products with discounted prices below ₹500, enabling focused analysis on low-cost items.

CONCLUSION

- THROUGH STRUCTURED SQL QUERIES, WE EFFECTIVELY ANALYZED AMAZON'S PRODUCT AND REVIEW DATASET TO EXTRACT MEANINGFUL INSIGHTS.
- WE EXPLORED VARIOUS DIMENSIONS SUCH AS PRICING, DISCOUNT PATTERNS, CUSTOMER SENTIMENT, AND PRODUCT CATEGORIZATION.
- THE DATA WAS FILTERED, SEGMENTED, AND INTERPRETED TO SUPPORT STRATEGIC DECISIONS LIKE PROMOTIONAL TARGETING, PRODUCT POSITIONING, AND INVENTORY PLANNING.
- THIS ANALYSIS SHOWCASES THE POWER OF SQL IN TRANSFORMING RAW DATA INTO ACTIONABLE INTELLIGENCE FOR BUSINESS GROWTH.
- THE APPROACH CAN BE SCALED AND ADAPTED FOR ONGOING ANALYTICS ACROSS LARGER DATASETS AND EVOLVING BUSINESS NEEDS.



The Amazon logo, featuring the word "amazon" in a dark blue, lowercase, sans-serif font. Below the text is a curved orange arrow pointing from the 'a' to the 'z'. The background is a light beige grid pattern with decorative orange dashed swirls in the top left and bottom right corners, and orange shapes with dark blue polka dots in the top right and bottom left corners.

amazon

The text "THANK YOU!" in a large, dark blue, uppercase, sans-serif font. A small orange leaf icon is positioned to the left of the text. The background is a light beige grid pattern with decorative orange dashed swirls in the top left and bottom right corners, and orange shapes with dark blue polka dots in the top right and bottom left corners.

THANK YOU!

For your attention.

Presented By:
Adnan Hafeez

